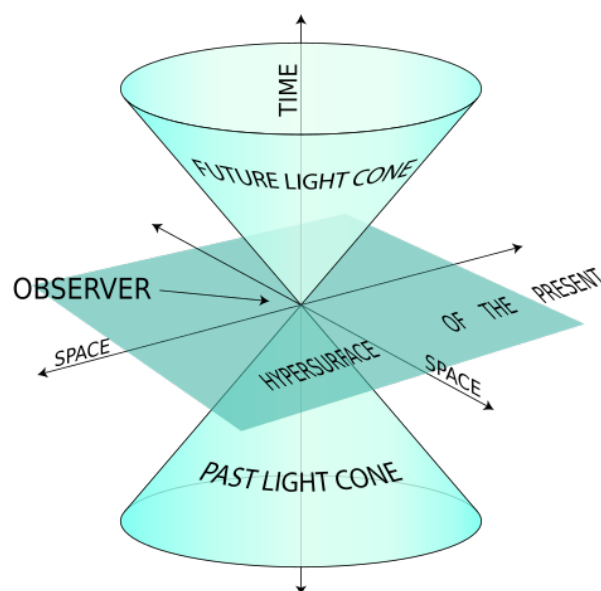

Notes on The Classical Theory of Fields

Volume 2 of the Course of Theoretical Physics, by *Landau & Lifshitz*



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Introduction

The Classical Theory of Fields [1], Volume 2 of the *Course of Theoretical Physics* by Landau and Lifshitz, presents an elegant and unified account of electrodynamics and general relativity. The work is devoted to the theory of electromagnetic and gravitational fields, building upon the framework of special relativity through variational principles to ensure generality, unity, and clarity of exposition.

In this framework, the special theory of relativity serves as the foundation for electrodynamics, ensuring consistency with the finite propagation speed of interactions and the relativistic structure of spacetime. From this starting point, the fundamental equations governing the electromagnetic field are derived, followed by the development of the general theory of relativity, culminating in a logically connected presentation of gravitational fields and the dynamics of spacetime.

The exposition is based on variational principles, constructing a Lagrangian (or action, which is a function of the Lagrangian) from which field equations and conservation laws emerge naturally, offering a cohesive and powerful methodological approach, according to most physicists' opinions.

Chapter 1

The Principle of Relativity

§1 Postulates

The speed of light is the same in all inertial reference frames.

Example : a man running at velocity v with a laser pointer activated in his hands, won't make the electromagnetic field propagate at $v_{\text{light}} = c + v$, because $v_{\text{light}} = c$ in all inertial frames no matter the velocity of the proper inertial frame relative to another.

An *inertial* reference system is a system of reference in which a freely moving body, proceeds with constant velocity.

If one such reference system has another moving at constant velocity relative to it, then that new moving reference frame is also *inertial*. The laws of physics are the same in all inertial systems of reference, thus all physical laws expressed in an inertial frame when written in terms of coordinate and time (which are the parameters that define a reference system) are of the same form when transformed from one inertial reference system to another.

There exists a maximum velocity of propagation of interaction, that is c , the speed of light. The propagation of such an interaction from a particle to another is called a *signal*, and it *informs* the second particle of the changes the first has experienced.

§2 Intervals

An *event* is a position associated with a time (the time is the time of the frame of reference of the position). The representation of events on a 4-dimensional space is that of points in said space, these points are called *world points*. These points form a line in the 4-dimensional space, called *world lines*.

Let us consider two inertial frames K and K' moving at constant velocity relative to each other, with coordinate axes X, Y, Z and X', Y', Z' respectively.

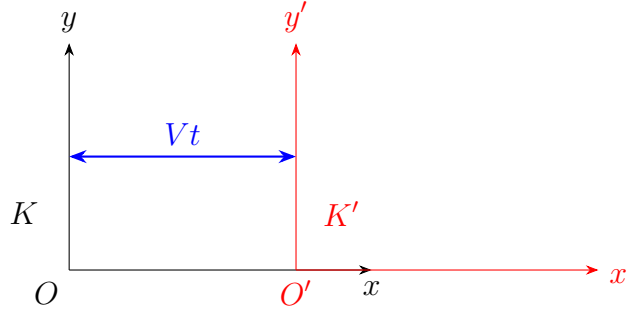


Figure 1.1: K and K' separated by Vt , the distance travelled in at time t by the frame moving at V

An event in K is characterized by (t, x, y, z) whereas an event in K' is characterized by (t', x', y', z') . Let there be a first event, starting point of a signal propagating at c , with coordinates (x_1, y_1, z_1) at t_1 , and a second event of coordinates (x_2, y_2, z_2) at t_2 . We see instantly that the distance it travelled is given by both equal equations :

$$v(t_2 - t_1) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} \quad (1.1)$$

As stated earlier, we consider the event to propagate at c , so $v = c$ and thus from equation (1.1), we square the two terms and state :

$$(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2 - c^2(t_2 - t_1)^2 = 0 \quad (1.2)$$

These two events in K' are "primed", so the first event is of coordinates (x'_1, y'_1, z'_1) at t'_1 , and the second event of coordinates (x'_2, y'_2, z'_2) at t'_2 , and since the speed of light is the same in all inertial frames, then $v' = c$ too, and finally by the same reasoning we have :

$$(x'_2 - x'_1)^2 + (y'_2 - y'_1)^2 + (z'_2 - z'_1)^2 - c^2(t'_2 - t'_1)^2 = 0 \quad (1.3)$$

We will write :

$$s_{12} = [c^2(t_2 - t_1)^2 - (x_2 - x_1)^2 - (y_2 - y_1)^2 - (z_2 - z_1)^2]^{\frac{1}{2}} \quad (1.4)$$

$$s'_{12} = [c^2(t'_2 - t'_1)^2 - (x'_2 - x'_1)^2 - (y'_2 - y'_1)^2 - (z'_2 - z'_1)^2]^{\frac{1}{2}} \quad (1.5)$$

s_{12} and s'_{12} are the intervals between these two events in K and K' respectively.

It is now obvious that for signals propagating at the speed of light, c , the intervals are all equal to 0 independently of the reference system.

If the two events are infinitesimally close, we have :

$$ds^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2.$$

And for $v = c$, $ds = ds' = 0$. Since intervals have the same units, we can conjecture that :

$$ds = a(V) ds' \quad (1.6)$$

With $a(V)$ a factor with no dimensions.

We introduce a factor $a(V)$ relating the intervals in two frames. By symmetry and isotropy of space, this factor can only depend on the magnitude of the relative velocity, not its direction.

This last argument allows to write the following equations :

$$ds' = a(-V)ds = a(V)ds \quad (1.7)$$

And using equation (1.6) and (1.7) we have :

$$ds = a(V)ds' = \frac{ds'}{a(V)}$$

Which leaves us with :

$$a(V) = \pm 1$$

To choose between signs, we consider the case where $v = 0$:

$$ds = ds',$$

since all lengths are equal to their equivalent in the other frame, so we must have :

Invariance of intervals : $ds = ds'$, in all reference frames.
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(1.8)

Let us introduce the notation :

$$t_2 - t_1 = t_{12} \quad \text{et} \quad (x'_2 - x'_1)^2 + (y'_2 - y'_1)^2 + (z'_2 - z'_1)^2 = l_{12}^2 \quad (1.9)$$

Let's talk about the types of intervals possible.

If we consider two events that happen at the same position in the K' reference system, we have :

$$l'_{12} = 0$$

Leading to the equality (1.8) becoming :

$$s_{12}^2 = c^2 t_{12}^2 - l_{12}^2 = c^2 t_{12}^2 > 0 \quad (1.10)$$

This kind of interval with $s_{12}^2 > 0 \rightarrow s_{12} \in \mathbb{R}$ is called a *timelike* interval. For a timelike interval there exists a frame in which the two events occur at the same spatial location.

Let me introduce another type of interval. Let us consider two inertial reference frames, K and K' , let us consider two events that happen simultaneously in K' . The interval between event 1 and 2, is given by :

$$s_{12}^2 = s_{12}'^2 \quad (1.11)$$

$$s_{12}^2 = c^2 t_{12}'^2 - l_{12}'^2 \quad (1.12)$$

$$s_{12}^2 = -l_{12}'^2 \quad (1.13)$$

$$s_{12} = \sqrt{-l_{12}'^2} \quad (1.14)$$

$$s_{12} = i l_{12}' \in \mathbb{C} \quad (1.15)$$

So evidently the interval is imaginary, and we call such intervals *spacelike* intervals.

The last type of interval is the one we mentioned first, the *lightlike* interval, in which $ds = 0$ and that represents motion at constant velocity c .

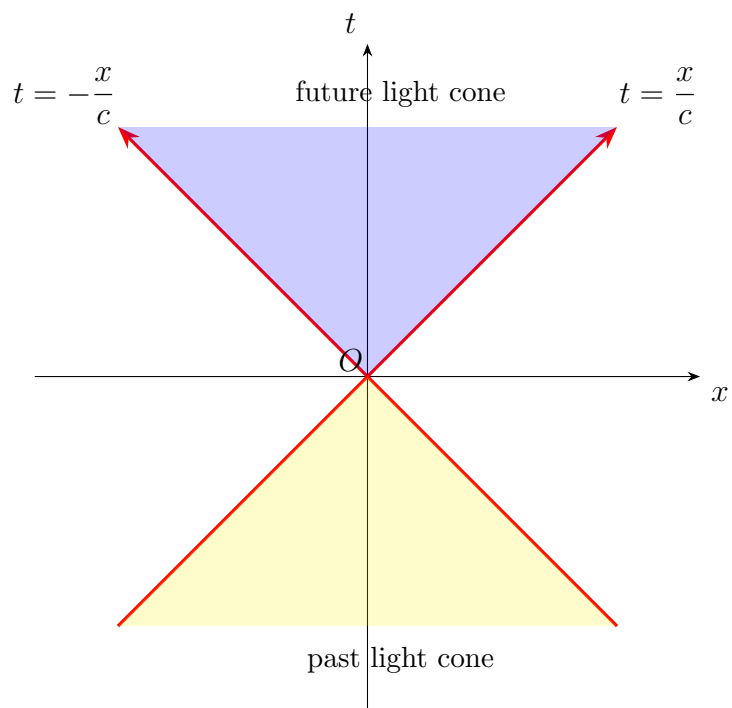


Figure 1.2: Light cone of the observer O

§3 Proper Time

Let us introduce the *proper frame*, associated to a timelike interval. This frame is at rest relative to the object whose proper time is measured. This frame is called the *proper frame* and the time elapsed in this frame is called the *proper time*, which will be written $d\tau$ by the end of me finding it's expression:

$$c^2 dt'^2 = c^2 dt^2 - dl^2 \quad (1.16)$$

$$dt'^2 = dt^2 - \frac{dl^2}{c^2} \quad (1.17)$$

$$dt' = \sqrt{dt^2 - \frac{dl^2}{c^2}} \quad (1.18)$$

$$dt' = dt \sqrt{1 - \frac{dl^2}{c^2 dt^2}} \quad (1.19)$$

$$d\tau = dt \sqrt{1 - \frac{v^2}{c^2}} \quad (1.20)$$

As the postulate indicates that $c \geq v$, $1 \geq 1 - \frac{v^2}{c^2}$, therefore : $1 \geq \sqrt{1 - \frac{v^2}{c^2}}$, and that ends up as saying that :

$$d\tau \leq dt \quad (1.21)$$

In a frame moving at a velocity v relative to another, an observer sees a clock moving relative to them as ticking more slowly than a clock at rest in their own frame. That is, if we have a clock in K and another in K' , each observer sees the other's clock running slower than their own. This is the notion of *time dilation*¹ in special relativity.

¹On the twin paradox's validity : simultaneity isn't absolute, and not only time dilation but also length contraction take part in the process, acceleration isn't the key. Acceleration is absolute, since it's effects are equivalent to a Force which too is absolute.

Bibliography

- [1] L. D. Landau and E. M. Lifshitz. *The Classical Theory of Fields*, volume 2 of *Course of Theoretical Physics*. Pergamon Press, 4th edition, 1975.